HOMEWORK 1: NEURAL NETWORKS & INTRODUCTION TO CUDA C

Q1: What are autoregressive models? Provide real-world examples. (5 points)

Ans: Autoregression is a powerful statistical technique where in past values of a variable are used to predict the future values. The use of the term "auto" implies that the variable regresses on itself, i.e. it uses the past values as predictors of its future values. The models that work on the concept of autoregression are autoregressive models.
These models work on the following process:
Input and probability: Considering the initial set of tokens as a prompt, it predicts the probability of the next token that is likely to occur in the output.
Sampling: It samples from the list of contenders to choose the next token.
Iteration: The chosen token is added to the end of the input. The new result is then fed back into the model to predict the next token.
Self Attention and end of loop: Masked self attention is used to "mask" the tokens beyond what the model is currently trying to predict. Once an end-of-sequence token is predicted, the loop stops and the final output is provided.
Some real world examples include:

1. Large Language Models: The classic LLMs generate human-like text when they read the input sequence from left to right, while predicting the next token(word) based on all preceding words.
2. Temperature and Weather forecasting: The weather for the next day can be predicted based on the weighted average of the current and/or preceding days.
3. Biology and Drug discovery: Instead of writing words, these models are prompted with specific evolutionary traits or structural requirements. The AR engine then generates entirely new, synthetic amino acid sequences one letter at a time. This allows scientists to design novel, functional enzymes and targeted therapeutics that do not exist in nature.
4. Interactive video generation and simulation: The architecture generates video frames sequentially or processes historical frames to predict how a scene, physics environment, or lighting shift will unfold across the next frame. This acts as a predictive physics simulator for virtual environments.
5. Financial and economic projections: Economists model consumer price index, inflation, or gross domestic product (GDP). Standard AR and ARIMA setups use historical price levels or financial asset returns to project future baseline economic trends.
6. Computer programming and software development: When a developer writes a function signature, the AR model predicts the next logical characters, syntax structures, or entire blocks of code based on the context of the open file and imported libraries. It functions exactly like an advanced autocomplete engine tuned for programming logic.